WEEK 2 · DAY 5 · OVERVIEW

Case Study

Three Years of Curve and Spreads



Created by **Jeff Lenamon**

ABOUT 45 MIN WITH EXERCISES / 10 SECTIONS / 4 EXERCISES

This course is educational content created in a personal capacity. Nothing here is investment advice or a recommendation to buy or sell any security. All portfolio data is synthetic.

Problem Definition

Week 2 ends with one table. For any date and any set of series, one row per series: the level, the move over a day, a week and a month in bp, and where it sits in its last year, tied to Excel and built from nothing after that date. Put it in marketdata with a test that proves it, and tag the week. Then use it: describe the curve and our two synthetic spreads over the three years FRED held of ICE's series when we checked on 4 October 2026, 3 October 2023 to 2 October 2026, on one four-panel page. With a key, show me ICE's own investment grade and high yield series arrived, in rows and dates only. Describe; no view on where anything goes.

ABOUT 45 MIN WITH EXERCISES / 10 SECTIONS / 4 EXERCISES

Your Route Through Day 5

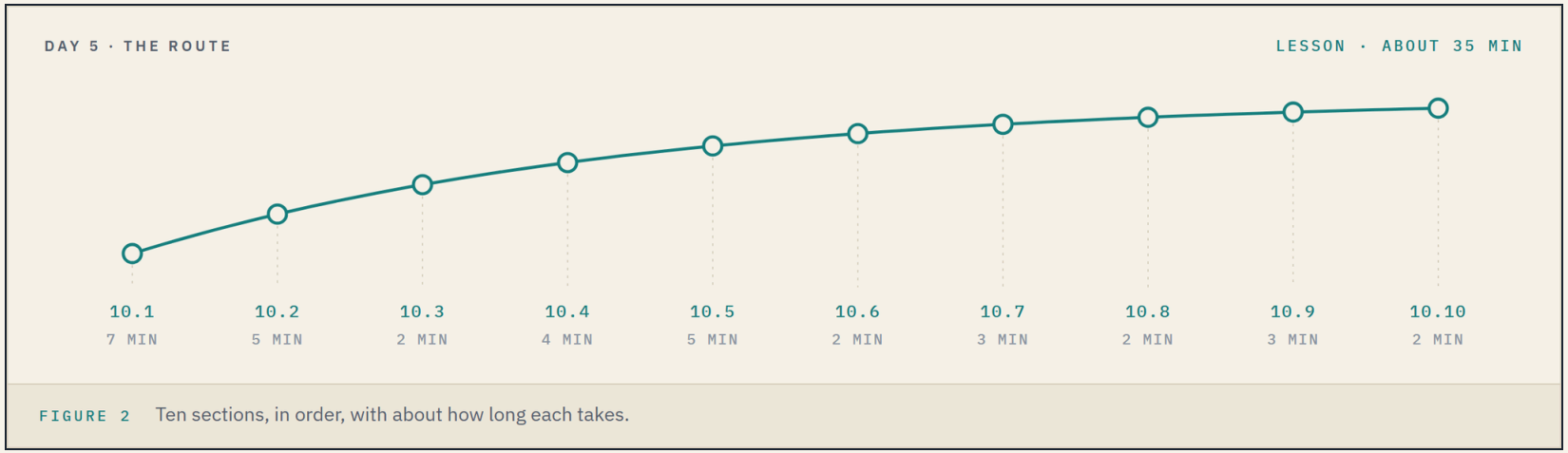


FIGURE 2 Ten sections, in order, with about how long each takes.

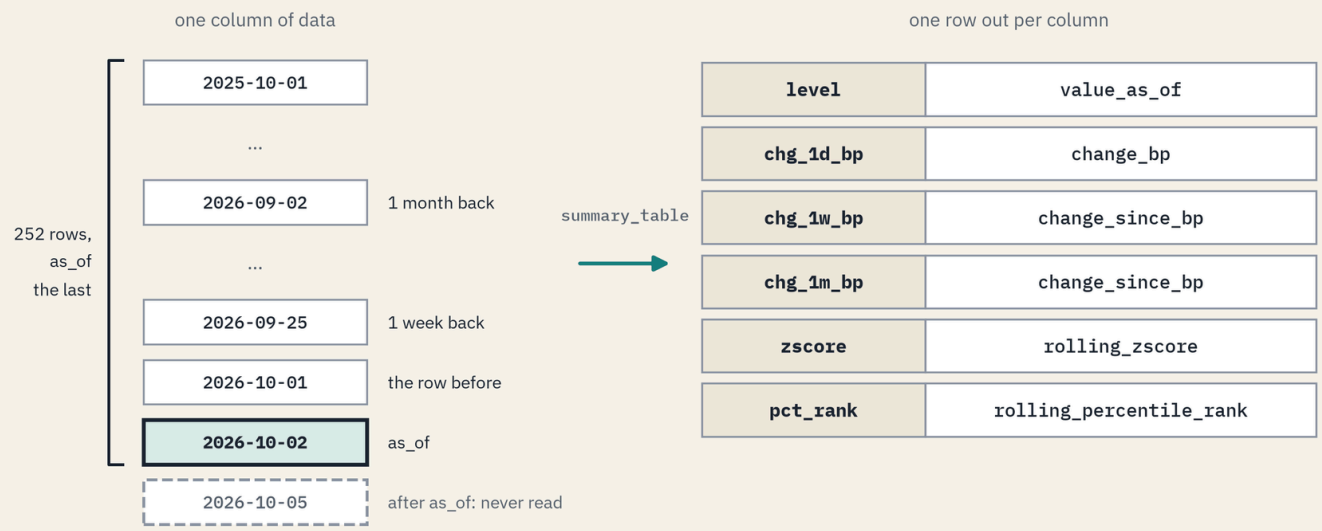
	SECTION	ABOUT
10.1	One Row per Series: `summary_table`	7 min
10.2	Nothing After the As-Of Date	5 min
10.3	Tag the Week	2 min
10.4	Live: The Same Table From FRED, and ICE in Outline	4 min
10.5	Three Years of Levels and Daily Moves	5 min

	SECTION	ABOUT
10.6	Volatility Regimes	2 min
10.7	Where Each Series Sits: The First Day and the Last	3 min
10.8	The Shape of the Curve	2 min
10.9	The Synthetic Spreads	3 min
10.10	One Page, Four Panels	2 min

Time. About 35 minutes for the lesson, then about 12 for the four exercises.

10.1 One Row per Series: summary_table

One row per series: every measure as of one date, from rows on or before it



Units are given, never guessed: a yield in percent has its changes times 100, a series in bp does not.

FIGURE 3 One column of rows to 2026-10-02 becomes one row of level, 1d, 1w, 1m, z-score and rank.

ONE ROW PER SERIES

SERIES	UNIT	LEVEL	CHG_1D_BP
10 Yr	pct	5.28	4.00
2s10s	bp	45.00	-1.00

`summary_table` **puts the 10 Yr and 2s10s on 2 October 2026 in one row each, equal to Excel's reference row within 4.6e-13.**

10.2 Nothing After the As-Of Date

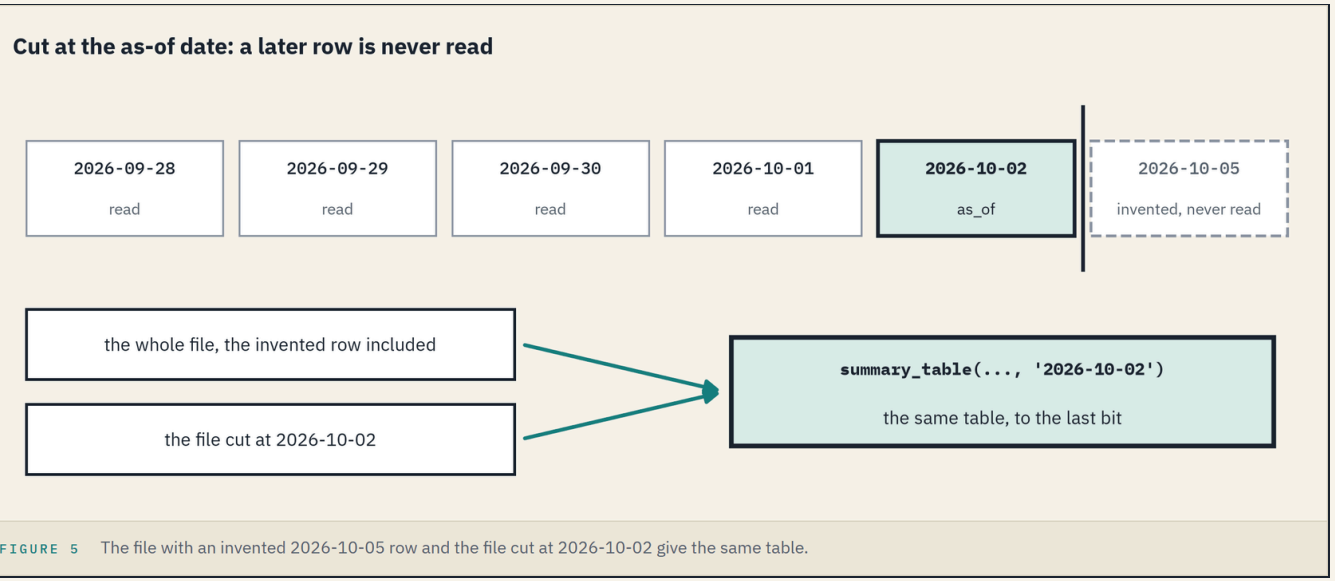


FIGURE 5 The file with an invented 2026-10-05 row and the file cut at 2026-10-02 give the same table.

A MADE-UP ROW

```
10 Yr  2s10s
date
2026-10-01   5.24   46.0
2026-10-02   5.28   45.0
2026-10-05   9.99  500.0
```

THE TABLE DOES NOT MOVE

```
seen = md.summary_table(pair_plus, '2026-10-05', pair_units).loc['10 Yr', 'chg_1d_bp']
print(f'As of 2026-10-02 nothing moved; as of 2026-10-05 the row is read: {seen:+.1f} bp')
```

```
As of 2026-10-02 nothing moved; as of 2026-10-05 the row is read: +471.0 bp
```

10.3 Tag the Week

COMMIT

```
M marketdata.py
 M test_marketdata.py
 marketdata.py      | 40 ++++++++++++++++++++++++++++++++++++++++
 test_marketdata.py | 35 ++++++++++++++++++++++++++++++++++
 2 files changed, 75 insertions(+)
```

TAG THE WEEK

```
!git -C "{work}" tag week2
```

```
51b40fc (HEAD -> main, tag: week2) Notebook 10: summary_table
19940f9 Notebook 09: slope_bp, butterfly_bp, fly_weights, and slopes_2026.csv
de7069f Notebook 08: rolling_zscore and rolling_percentile_rank
```

week2 **names the module as week 2 ends; between the tags lie notebooks 06 to 10, a data file among them. Now the function earns its keep.**

10.4 Live: The Same Table From FRED, and ICE in Outline

THE SAME TABLE FROM FRED

```
live_2y = md.fred_series('DGS2', fred_key, start='2023-10-03')
live_10y = md.fred_series('DGS10', fred_key, start='2023-10-03')
live_curve = pd.DataFrame({'2 Yr': live_2y, '10 Yr': live_10y}).dropna()
live_pair = pd.DataFrame({'10 Yr': live_curve['10 Yr'],
live_table = md.summary_table(live_pair, '2026-10-02', pair_units)
pd.testing.assert_frame_equal(live_table, summary, check_exact=False, rtol=0, atol=1e-9)
```

ICE'S TWO SERIES, IN MEMORY

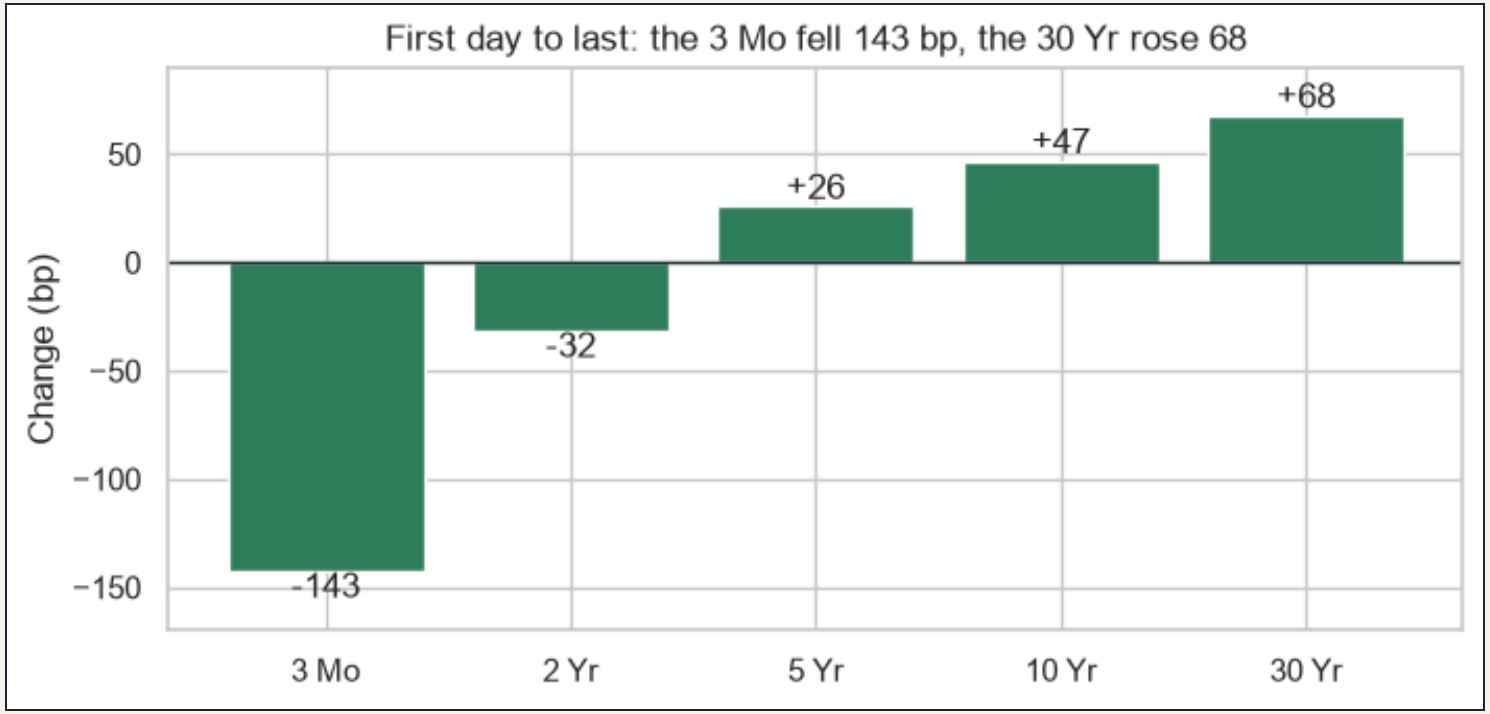
```
live_ice = md.fred_series(live_id, fred_key, start='2023-10-03')
live_dots = live_ice.isna().sum()
live_weekend = (live_ice.index.dayofweek >= 5).sum()
assert live_ice.index.is_unique and live_ice.index.is_monotonic_increasing, live_id
```

THEN A STUDENT SEES

The first cell prints FRED's row count, its last date and the check line, since FRED's **DGS** series are Treasury's own numbers.

With a key the same table comes from FRED, and ICE's two series arrive as rows and dates. Every number from here on is the Treasury file or the synthetic spreads.

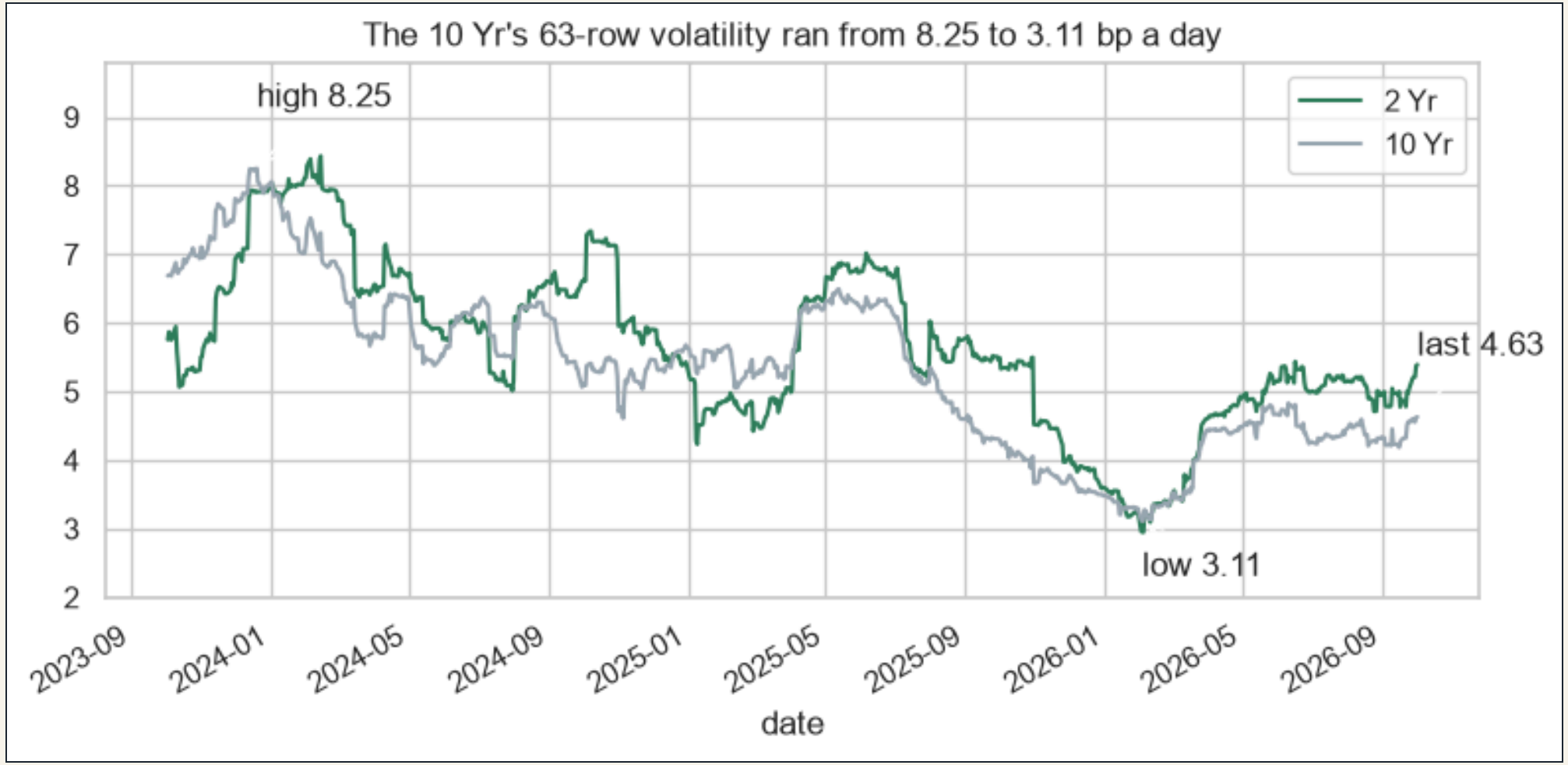
10.5 Three Years of Levels and Daily Moves



THE CHART'S CODE

```
bars = ax.bar(levels.index, levels['moved_bp'])
ax.bar_label(bars, fmt='%+.0f')
ax.set_title('First day to last: the 3 Mo fell 143 bp, the 30 Yr rose 68')
```

10.6 Volatility Regimes



10.7 Where Each Series Sits: The First Day and the Last

SERIES	UNIT	LEVEL	CHG_1D_BP	CHG_1W_BP	CHG_1M_BP	ZSCORE	PCT_RANK
3 Mo	pct	4.19	2.00	-5.00	27.00	2.58	98.02
2 Yr	pct	4.83	5.00	2.00	44.00	2.51	98.02
5 Yr	pct	5.06	5.00	8.00	52.00	2.80	99.60
10 Yr	pct	5.28	4.00	11.00	49.00	3.08	99.60
30 Yr	pct	5.63	2.00	14.00	36.00	3.08	99.60
2s10s	bp	45.00	-1.00	9.00	5.00	-0.51	28.17
5s30s	bp	57.00	-3.00	6.00	-16.00	-2.07	5.95
3m10y	bp	109.00	2.00	16.00	22.00	2.07	100.00
2s5s10s	bp	1.00	1.00	3.00	11.00	2.70	98.81
5s10s30s	bp	-13.00	1.00	0.00	10.00	2.58	100.00
ig_spread_bp	bp	97.00	-1.00	1.10	1.90	0.22	61.11
hy_spread_bp	bp	328.70	-1.30	11.20	0.10	0.90	73.81

10.8 The Shape of the Curve

THE LAST DAY BELOW ZERO

```
for name in ['2s10s', '3m10y']:
    print(f'{name}: last below zero on {shape[shape[name] < 0].index[-1]:%Y-%m-%d}')
```

```
2s10s: last below zero on 2024-09-05
3m10y: last below zero on 2025-10-16
```

EXCEL SIDE BY SIDE

JOB	PYTHON	EXCEL	RESULT
2s10s days below zero	`signs.loc['2s10s', 'below_zero']`	`=COUNTIF(J3:J752,"<0")`	227

10.9 The Synthetic Spreads

RANGE AND GAP

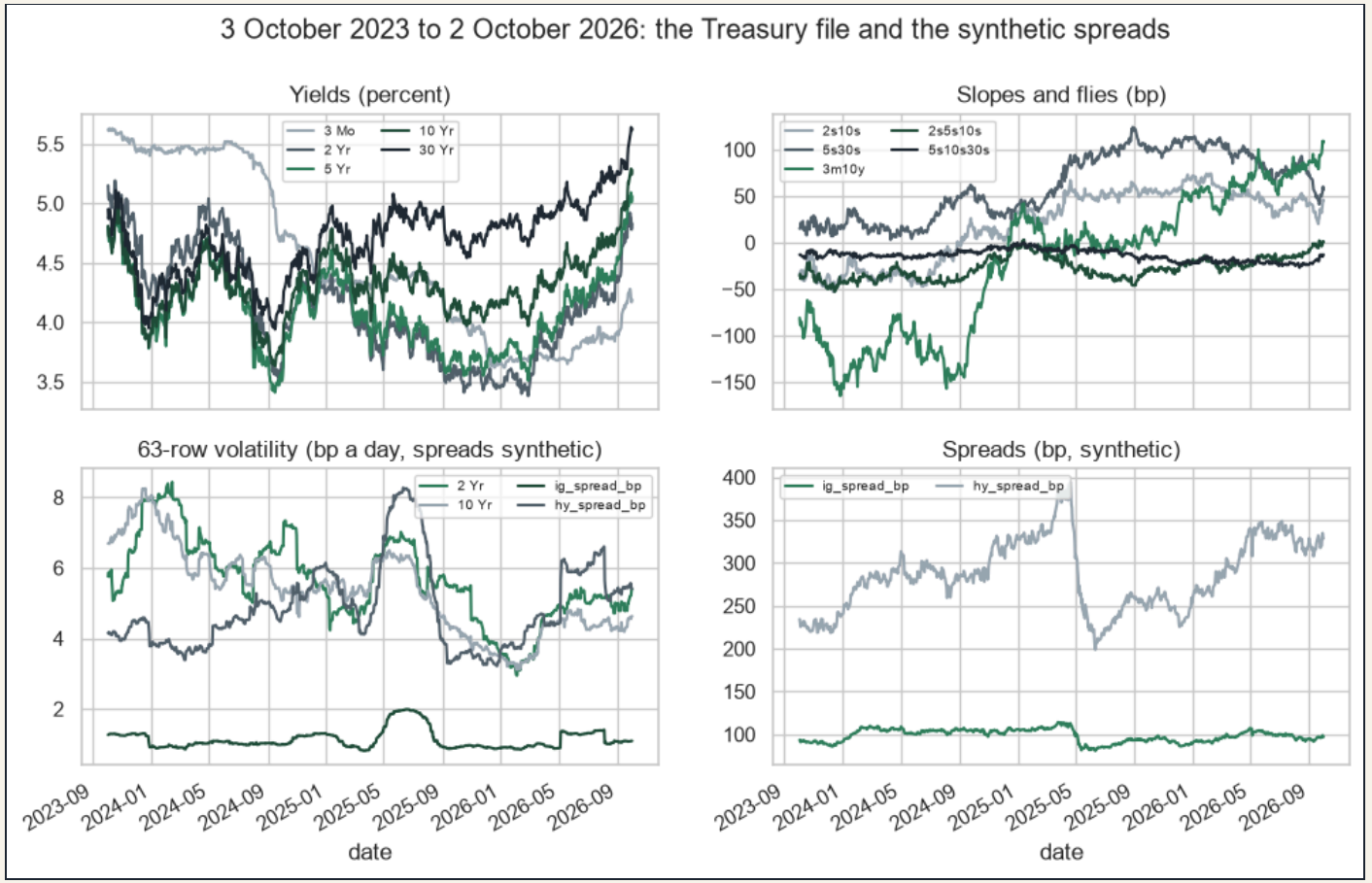
```
synthetic = three_years[['ig_spread_bp', 'hy_spread_bp']]
gap = synthetic['hy_spread_bp'] - synthetic['ig_spread_bp']
```

```
first   last    low   high
ig_spread_bp   93.5   97.0   80.7  114.2
hy_spread_bp  234.3  328.7  198.7  394.3
gap           140.8  231.7  118.0  281.4
```

EXCEL SIDE BY SIDE

JOB	PYTHON	EXCEL	RESULT
Narrowest gap	`gap.min()`	`=MIN(K3:K752)`	117.99999999999999
Widest gap	`gap.max()`	`=MAX(K3:K752)`	281.4

10.10 One Page, Four Panels



One page, four panels, six observations, every number from a table above: the note is answered.

You Can Now

- Build one row per series of every measure as of one date with `summary_table`.
- Prove a table reads nothing after its date, with an invented row and a test.
- Tie a table to an Excel sheet built with **XMATCH**, **XLOOKUP** and **OFFSET**.
- Tag the end of a week and read what changed between two tags.
- Describe three years of a curve and two spreads in numbers and dates, with no forecast.

NEXT

WEEK 3 · DAY 1

SQL With sqlite3

The book in a database file: SQL with Python's own sqlite3, every query beside the pandas you trust, and an assert that proves the two return the same rows.